UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Ingegneria Gestionale, dell'Informazione e della Produzione

Spatio-Temporal Analysis of NO_2 and CO in Beijing

Statistics for High Dimensional Data Course Project

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03 Feature Engineering and Selection

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Introduction

01

NO_2 and CO

Nitrogen Dioxide (NO_2)

- Produced by high-temperature combustion (heating, traffic, industry)
- Higher density than air → concentrates near ground level

Carbon Monoxide (CO)

- Generated by incomplete combustion processes
- Sources: both natural (wildfires) and anthropogenic (traffic, heating)

- *Both pollutants are regulated by WHO and continuously monitoring across urban and suburban areas.*
- *Meteorological variables (wind, temperature, pressure, humidity) critically influence transport and dispersion of both pollutants*

Data and Exploratory Analysis

02

Dataset Overview

Description

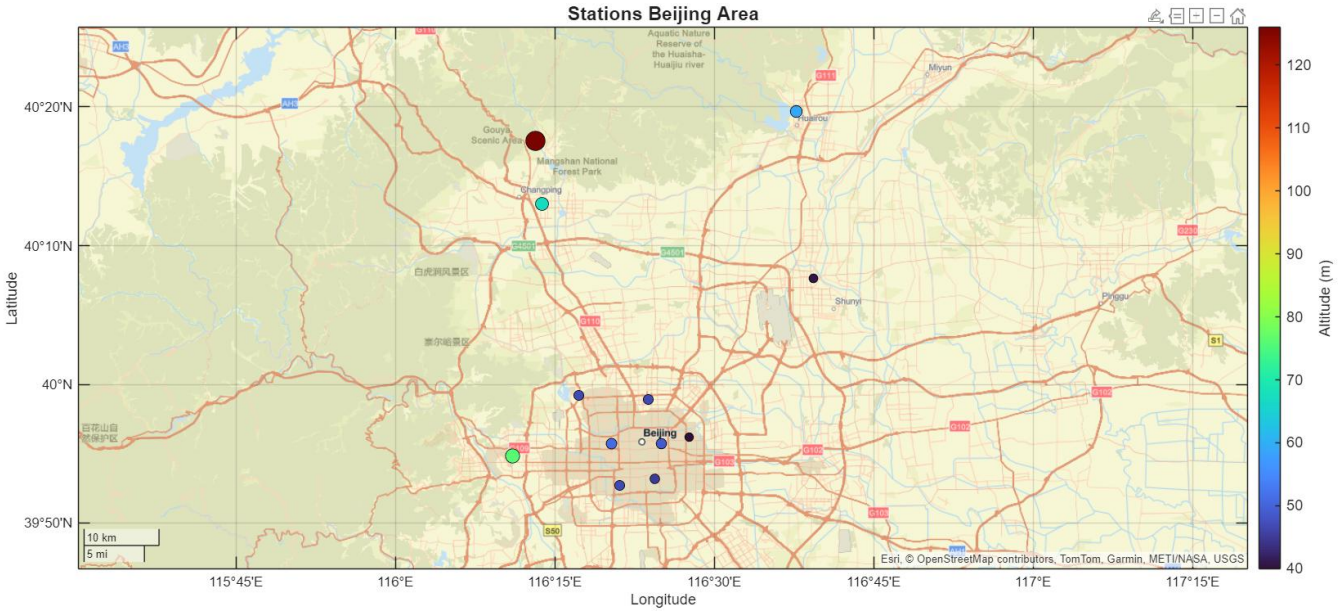
Source	Kaggle – Beijing Multi-Site Air Quality Data
Stations	12 monitoring stations in Beijing Area
Variables	NO_2, CO, PM2.5, PM10, SO_2, O_3 TEMP, PRES, DEWP, RAIN, WSPM, wind direction, timestep and station

Preprocessing

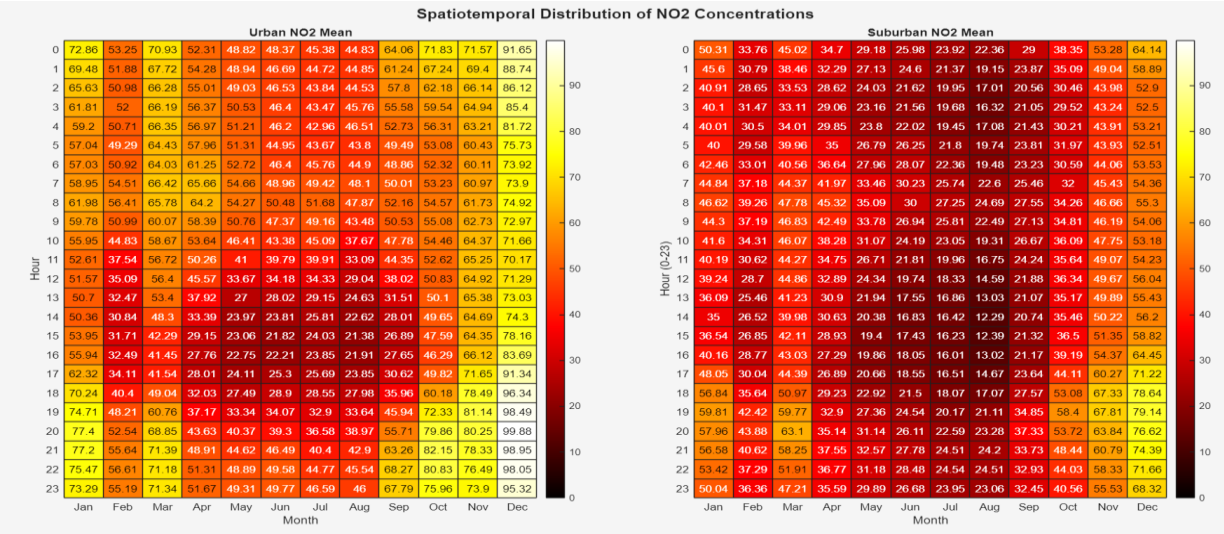
Observations	195,180 hourly records after preprocessing
Missing data	NO_2 and CO: 0.9%–3.7% per station → no imputation applied TEMP, PRES, DEWP, RAIN, WSPM → Linear interpolation
Geo data	Latitude, longitude, altitude added from external sources

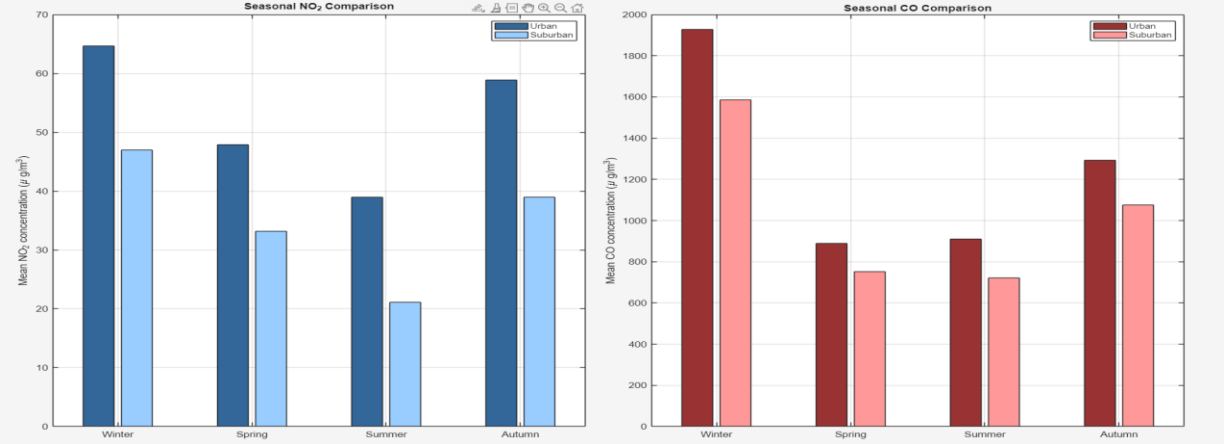
Stations

Station	Lon	Lat	Alt(m)	Zone
Aotizhongxin	116.397	39.982	47	Urban
Changping	116.230	40.217	67	Suburban
Dingling	116.220	40.292	126	Suburban
Dongsi	116.417	39.929	49	Urban
Guanyuan	116.339	39.929	51	Urban
Gucheng	116.184	39.914	76	Urban
Huairou	116.628	40.328	59	Suburban
Nongzhanguan	116.461	39.937	40	Urban
Shunyi	116.655	40.127	41	Suburban
Tiantan	116.407	39.886	45	Urban
Wanliu	116.287	39.987	47	Urban
Wanshouxigong	116.352	39.878	47	Urban



Key EDA Findings





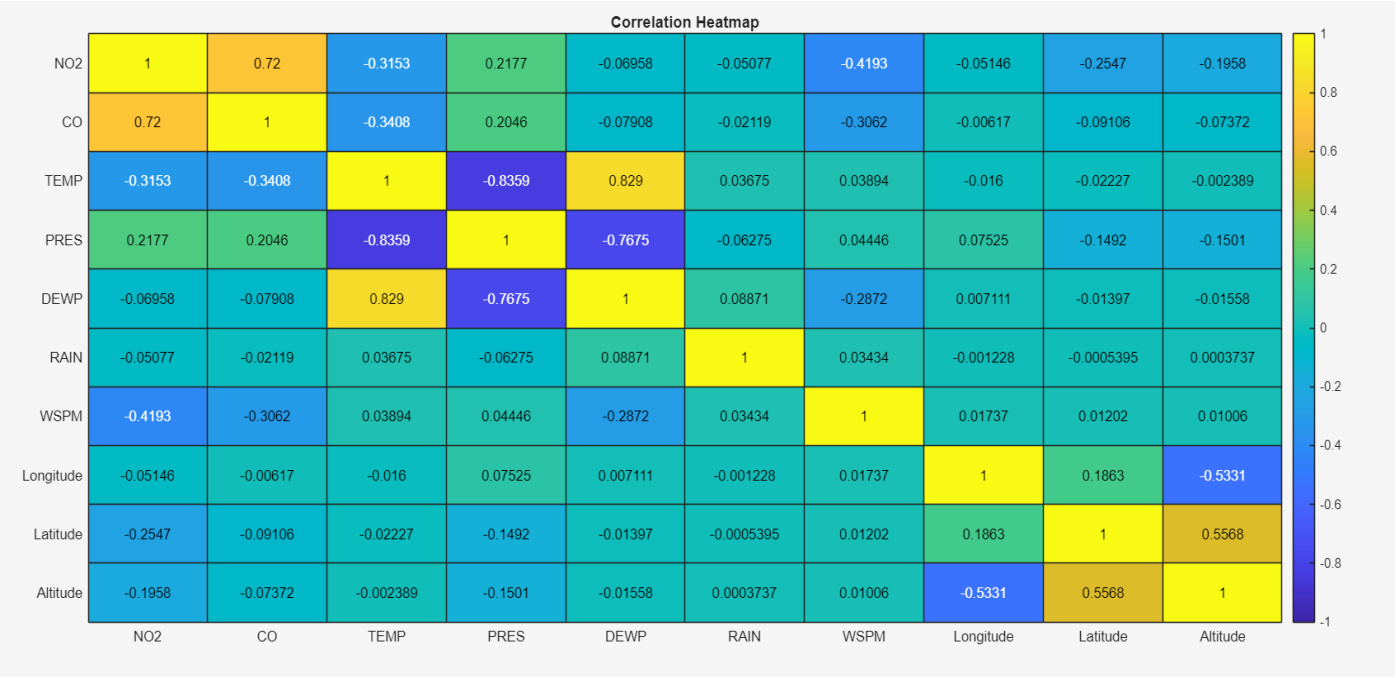
Urban vs Suburban

8 urban + 4 suburban stations with distinct pollutant dynamics.

Seasonal Pattern

Clear winter peak / summer trough. Linked to increased heating and reduced atmospheric dispersion.

Key EDA Findings



Multicollinearity

Strong correlation among TEMP, DEWP and PRES — must be addressed before modelling.

Strong Correlation between NO2 and CO

NO_2 and CO are strongly correlated (0.72), probably because they share many common sources

Feature Engineering and Selection

03

Feature Engineering

Seasonal Variables (sea_sin, sea_cos)

```
sea_sin = sin(2π(month−1)/12)
sea_cos = cos(2π(month−1)/12)
```

Continuous encoding of seasonality using sine/cosine — avoids discontinuity between December and January

Cumulative Rainfall (RAIN_24h)

```
RAIN_24h = movsum(RAIN, [23, 0])
```

Accumulated precipitation over previous 24 hours — more physically meaningful than instantaneous rainfall

Relative Humidity (Magnus-Tetens)

```
RH = 100 · exp(aTd/(b+Td) − aT/(b+T))
  a = 17.625,  b = 243.04
```

Single indicator combining TEMP and DEWP — reduces multicollinearity while preserving physical information

Binary Temporal Indicators

```
is_working_hour : 1 if 09:00−18:00
is_weekend : 1 if Saturday or Sunday
```

Feature Selection - Final Covariate Set

Variables Removed

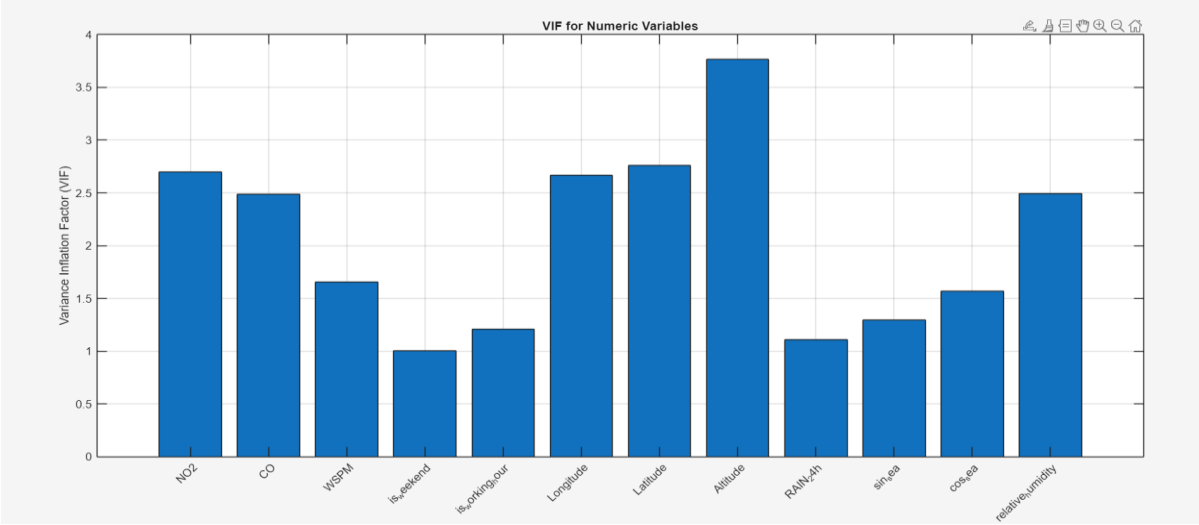
- TEMP – high correlation with DEWP (replaced by RH)
- DEWP – combined with TEMP into relative_humidity
- PRES – strong correlation with sea_cos
- RAIN – replaced by cumulative RAIN_24h

Final Covariate Set (10 variables)

- relative_humidity · WSPM · RAIN_24h
- latitude · longitude · altitude
- sea_sin · sea_cos · is_weekend · is_working_hour

VIF Analysis

Variance Inflation Factor (VIF) was computed to quantify multicollinearity. All variables in the final set showed VIF < 4. The joint use of correlation analysis and VIF ensured a well-conditioned predictor set — balancing spatial, temporal, and meteorological information.



Methodology

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Modeling Strategy

M1	M2	M3	M4
Linear Regression	**Univariate HDGM**	**Univariate HDGM**	**Bivariate HDGM**
Baseline — no spatio-temporal structure 10 covariates	Spatial covariates only (lat, lon, altitude)	Full set: spatial + temporal + meteorological (10 vars)	Joint NO_2 and CO modeling. Reduced covariate set (M3-based)

Validation strategy: Leave-One-Gauge-Out Cross-Validation (LOGOCV) — each of the 12 stations is iteratively removed and its observations predicted

Hidden Dynamic Geostatistical Model - Formulation

riate HDGM (M2, M3)

Observation equation:

$$y(s,t) = X(s,t)\beta + az(s,t) + \epsilon(s,t)$$

Latent state (Markovian):

$$z(s,t) = gz(s,t-1) + \eta(s,t)$$

Innovation (Gaussian Process):

$$\eta(s,t) \sim \mathcal{GP}(0, \rho(||s-s'||;\theta))$$

Measurement noise:

$$\epsilon(s,t) \sim \mathcal{N}(0, \sigma_\epsilon^2)$$

Parameter set:

$$\Psi = \{\beta, a, \sigma_\epsilon^2, \theta, g\}$$

Bivariate HDGM (M4)

Observation equation:

$$y(s,t) = X(s,t)\beta + az(s,t) + \epsilon(s,t)$$

Latent state (Markovian):

$$z(s,t) = Gz(s,t-1) + \eta(s,t)$$

Innovation (p-variate Gaussian Process):

$$\eta(s,t) \sim \mathcal{GP}_p(0, V\rho(||s-s'||;\theta))$$

Measurement noise:

$$\epsilon(s,t) \sim \mathcal{N}_p(0, \Sigma_\epsilon)$$

Parameter set:

$$\Psi = \{\beta, \Sigma_\epsilon, \theta, V, G\}$$

Results and Validation

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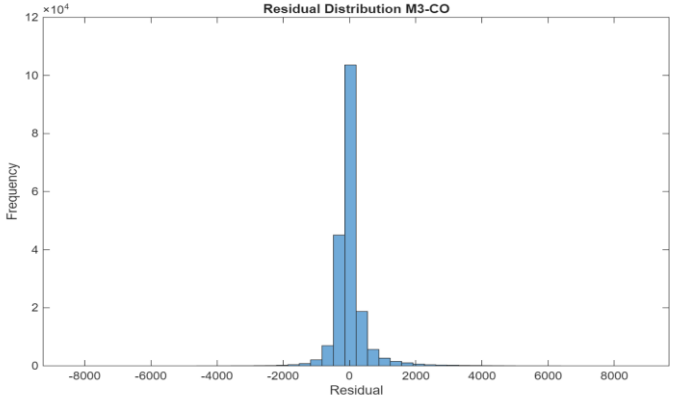
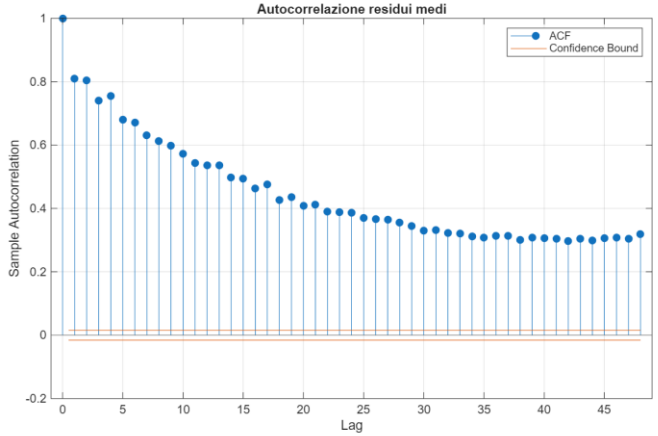
Model Performance Comparison – LOGOCV Metrics

Model	RMSE (NO_2)	R^2 (NO_2)	ACF(1) (NO_2)	RMSE (CO)	R^2 (CO)	ACF(1) (CO)
M1 – Linear Regression	26.46	0.32	0.85	964.19	0.39	0.89
M2 – HDGM (spatial only)	17.79	0.63	0.79	529.65	0.79	0.74
M3 – HDGM (full covariates)	17.34	0.65	0.78	510.15	0.81	0.72

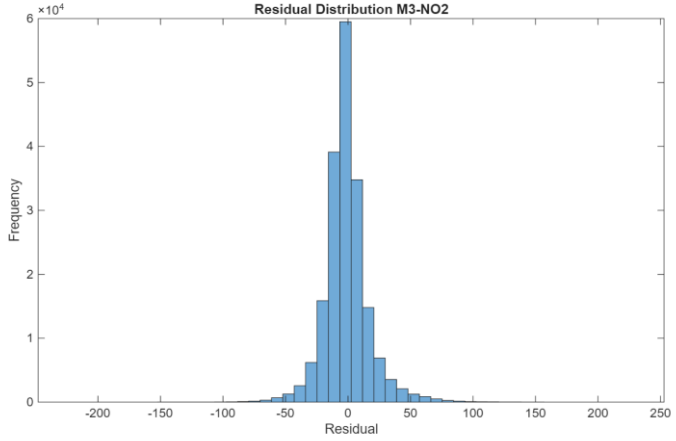
Model Performance Comparison – M3 Residuals Problem

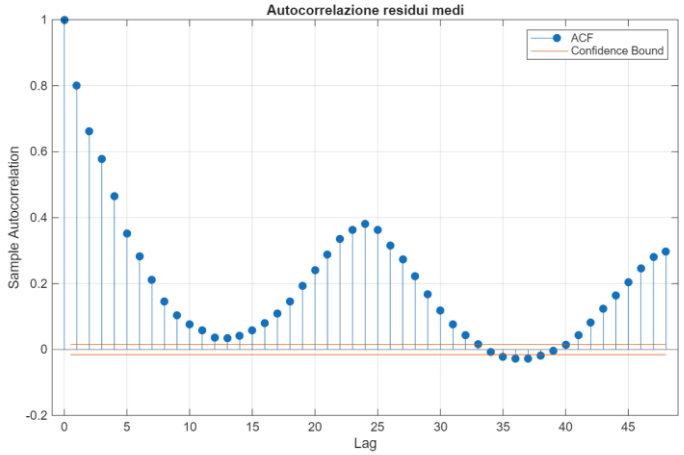
M3-CO





M3-NO_2





M3 - Estimated Coefficients and t-Statistics

Covariate	β (NO_2)	|t| (NO_2)	β (CO)	|t| (CO)
relative_humidity	0.370	*71.00*	0.359	75.96
RAIN_24h	-0.119	*14.74*	-0.050	6.93
WSPM	-0.013	*6.93*	-0.005	2.82
sea_sin	0.047	*2.42*	-0.023	0.93
sea_cos	0.308	*15.85*	0.329	13.24
latitude	-0.129	*7.60*	-0.069	3.52
longitude	-0.224	*14.80*	-0.074	4.29
altitude	-0.321	*24.58*	-0.102	7.28
is_weekend	0.013	*0.94*	0.009	0.69
is_working_hour	-0.001	*0.33*	0.044	8.96

Model Performance Comparison - LOGOCV Metrics

Model	RMSE (NO_2)	R^2 (NO_2)	ACF(1) (NO_2)	RMSE (CO)	R^2 (CO)	ACF(1) (CO)
M1 - Linear Regression	26.46	0.32	0.85	964.19	0.39	0.89
M2 - HDGM (spatial only)	17.79	0.63	0.79	529.65	0.79	0.74
M3 - HDGM (full covariates)	17.34	0.65	0.78	510.15	0.81	0.72
M4 - Bivariate HDGM	17.65	0.64	0.76	493.43	0.830	0.69

Conclusions

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Key Conclusions

Linear model (M1) is inadequate

Low R^2 (0.32–0.39) and very high residual autocorrelation (ACF ≈ 0.86–0.89) confirm a simple regression cannot capture the spatio-temporal structure of atmospheric pollutants.

HDGM improves prediction accuracy

M2 alone reduces RMSE by ~33% for NO_2 and ~45% for CO relative to M1. Spatial structure is the dominant factor — even without meteorological covariates.

Additional covariates provide only marginal gains

The M2 → M3 improvement is limited. The bivariate M4 achieves the best CO performance (R^2=0.83, RMSE=493) with fewer predictors.

Temporal dependence remains partially unexplained

ACF(1) ≈ 0.69–0.86 in all models.

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Thanks for the attention!